Report for *Oil Price Uncertainty and Sovereign Spreads of Oil Importers*

1 Summary of the paper

The paper studies how the volatility of the oil price affects sovereign spreads of oil-importing emerging economies. Empirically, the author estimates a stochastic volatility process for WTI oil prices and documents, in a quarterly panel of 35 net oil importers with EMBI data over 1995–2019, a positive association between oil price volatility and changes in sovereign spreads. The author then embeds an AR(1) oil price process with stochastic volatility into an Eaton–Gersovitz sovereign default model with long-term debt, in which oil is an imported input in a CES technology. The main quantitative claims are that the spread's response to oil price changes is three times larger in high volatility states (a slope of 0.15 versus about 0.05), that borrowing becomes much more responsive to oil prices in high volatility states (a debt slope of 0.70 versus 0.16), and that removing oil price uncertainty yields welfare gains averaging 1.1% of consumption.

The question is well motivated, the empirical fact is worth documenting, and combining an estimated stochastic volatility process with a quantitative default model is a reasonable research design. Below, I separate my comments into "major" and "minor" categories. The major comments address issues that need to be resolved for the paper to be publishable, while the minor comments suggest improvements to clarity and presentation.

2 Major comments

1. About what uncertainty means in this paper. The paper's story is: higher volatility today makes agents expect higher oil prices tomorrow (Figure 4d), so they expect lower output, a weaker ability to repay, and more default risk, which raises spreads. The expected price level rises with volatility because of a standard property of log-normal variables: $E[p'] = \exp(m + \frac{1}{2}v)$, where m and v are the conditional mean and variance of $\log p'$, so more variance mechanically pushes up the expected level. The paper's statement that "an increase in uncertainty surrounding current oil prices implies heightened expectations regarding future oil prices" is exactly this mechanical property. In other words, part of what the model calls an effect of "uncertainty" is really an effect of a higher expected oil price. That is fine, I just want to highlight that this is different from a typical uncertainty shock. When σ rises, the distribution of tomorrow's price changes in two ways: its mean moves up, a first-moment effect through which the economy simply expects more expensive oil, and it gets

wider, the mean-preserving increase in dispersion that the uncertainty literature (for example Fernández-Villaverde et al. (2011)) calls an uncertainty shock. Note that the specification itself has been used in the literature, but the wedge between the two channels is consequential here. In Fernández-Villaverde et al. (2011) the volatility shock hits the interest rate in levels, so it is mean-preserving by construction. Where the literature applies stochastic volatility to logged variables, as with income in Seoane (2019), the induced wedge is tiny and raises expected income, working against the measured uncertainty effect rather than with it. Here the wedge is large, given quarterly log price innovations above 10% and persistence of 0.98, and it raises the expected import cost, reinforcing the claimed effect; that is why the decomposition matters. To be clear about what the paper already does and what I am asking for: the full versus intermediate comparison in Section 6.2 shows the time-varying volatility matters, by shutting volatility shocks down entirely, and it is a sensible exercise for that question. It cannot show *why* volatility matters, because turning volatility shocks off removes the dispersion effect and the expected-price effect at the same time, and the comparison economies do not face the same average oil price (see also the next comment). An exercise that can separate those two channel is to subtract $\frac{1}{2}\text{Var}(\log p' \mid \sigma)$ from the conditional mean of $\log p'$, so that $E[p' \mid p, \sigma]$ no longer varies with σ and volatility states differ only in dispersion. With this decomposition, the results could be further interpreted. If the pure effect of uncertainty turns out to be small, I think it is fine. At least we know why volatility matters here.

2. Relatedly, the welfare comparisons mix two different things, less risk and cheaper oil. The comparison economies do not face the same average oil price. The basic model fixes the price at $e^{\bar{p}} \approx 1.13$, which is the median of the log-normal distribution, not its mean, so the full economy pays more for oil on average than the basic economy. The full and intermediate economies also differ: the full model's unconditional innovation variance is $E[\exp(2\sigma)]$, which exceeds the intermediate model's $\exp(2\mu)$ by convexity, and a more variable log price means a higher average price level. So every step of the full/intermediate/basic ladder (stochastic volatility, constant volatility, constant price) mixes "more risk" with "more expensive oil on average," and Figure 4d suggests these price-level differences are sizable. Part of the higher mean spread in the full model (4.0 versus 3.1 versus 2.5%), and part of the 1.1% welfare gain, may therefore simply reflect that the basic economy buys oil more cheaply, not that it faces less risk. The author could try to construct the comparison economies to match the full model's *average* price level, and also construct a comparison economy with only dispersion as suggested in the last comment.

3. In the model, a country in default keeps importing oil at the world price with no friction. The only direct cost of default is the endowment loss $h(y)$. In the data (paper's own motivating example, Sri Lanka's difficulty paying for fuel imports ahead of its 2022 default), and with the cited literature in which the loss of imported inputs is precisely what makes default costly

(Mendoza and Yue, 2012), it could be that the access to oil is disrupted during default times. The author should discuss this assumption; even better would be a robustness exercise in which default also disrupts oil imports, for example through a price wedge or partial exclusion from oil purchases during autarky.

4. Some claim of novelty is too strong as written, and the paper can do a better job in explaining what exactly is new relative to the closest papers. The paper states that "none of the existing research has addressed the impact of oil-related volatility shocks within the framework of sovereign default literature." This is too strong: the paper's own reference list contains Lopez-Martin et al. (2019), a sovereign default model with stochastic commodity revenues and hedging, cited but not discussed, and the paper's own footnote 1 acknowledges oil and default studies for exporters (Hamann et al. 2018; Esquivel 2020). The accurate, and still valuable, gap statement concerns oil importers and estimated stochastic volatility in oil prices specifically. The differentiation from the closest papers is also thin and could be explained in detail.

5. The model's headline state-dependent prediction could be tested directly in the data, which would strengthen the link between data and model. The claim is that the spread's sensitivity to oil price changes is three times larger in high volatility states. Its direct empirical analog is an interaction between the change in the oil price and a high-volatility indicator (or volatility itself) in equation (1). Also, the model implies that spreads of more oil-dependent economies should respond more strongly to oil price volatility. This is also testable by interacting volatility with the country-level oil import share (already computed from WITS for the calibration).

6. The effect in the model is much smaller than that in the data, and it would be helpful to discuss what governs its magnitude. The introduction translates the model's spread slopes of 0.05 and 0.15 into a 0.5 and a 1.5 basis point spread increase for a 10% oil price change, tiny movements against a mean spread of 410 basis points. The paper's own regressions suggest much bigger effects: the Table 1 coefficients of 0.14 to 0.19, of the percentage change of spreads on the change in the log oil price, translate into roughly 6 to 8 basis points per 10% oil price change at the mean spread. So the model delivers only a small fraction of the empirical response. A discussion of what determines this magnitude in the model would help the reader interpret the result and see what could close the gap. Also, the author could estimate regression (1) on model-simulated data, with volatility in the same units as the empirical measure, so that the model's coefficients can be compared with the empirical ones.

3 Minor comments

1. Notation: in equation (4) and in the constraint below equation (8), the bond price is written $q(b', y)$, but the schedule defined in equation (11) and used in the timing section depends on (b', y, p, σ). Also, for the CES aggregator in equation (3) the elasticity of substitution is $1/(1-\nu)$, not $1/\nu$ as stated in the text and labeled in Table 2; with $\nu = 0.5$ both happen to equal 2, so the calibration is unaffected, but the formula should be corrected.

2. Table 3 contains no oil-related moment: no correlation of the simulated spread with the oil price or its volatility, no volatility of the oil import share, and no comparison of the model's default frequency with the data. Adding such moments would be informative.

References

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